STATISTICS · EXPLORING DATA

Residuals & Influence Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Residuals & Influence Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If r = 0.98, must a straight line be a good model for the data? Explain.
   No. A curved relationship can have r near 1; check the residual plot for a pattern.
2. If study time changes from hours to minutes, which change: the slope, r, r^2?
   Only the slope (divided by 60). r and r^2 have no units and stay the same.

2 Explore use the tool

1. Study data: record the LSRL, r, r^2 and s.
   $\hat{y}$ = 51.97 + 3.706x; r = 0.9543; r^2 = 0.9107; s = 3.552 points.
2. Interpret the slope and r^2 in context.
   Each extra hour of study predicts 3.706 more points. About 91.1% of the variation in test scores is accounted for by the linear model on study time.
3. Age and height data: record r and describe the residual plot. Is a line appropriate?
   r ≈ 0.985, but the residuals are negative, then positive, then negative (curved). A line is not appropriate.
4. Use the extra point to find (a) an outlier that is not influential, (b) an influential point and (c) a high-leverage point that is not influential. Record coordinates.
   Answers vary, e.g., (a) (4, 30), (b) (12, 0), (c) (12, 96).

3 Record use the tool and fill in the table

One extra point added to the study data

Thresholds: outlier |residual| > 2s; high leverage x more than 2 SDs from $\bar{x}$; influential slope change ≥ 50%.

Extra point	Residual	Slope change	Outlier?	High leverage?	Influential?
(4, 30)	−33.96	+1.6%	yes	no	no
(12, 0)	−55.48	−125.2%	yes	yes	yes
(12, 96)	−0.26	−0.6%	no	yes	no
(5, 72)	+1.37	+0.3%	no	no	no

4 Explain what did you discover?

1. A residual = actual y − predicted $\hat{y}$.
2. A curved pattern in the residual plot means a line is not an appropriate model.
3. r^2 is the fraction of the variation in y that is accounted for by the linear model on x.
4. A point is influential if removing it changes the regression line a lot.

5. Why can a point with high leverage fail to be influential? Use an example from the lab.
   If its y-value is close to what the line predicts, it pulls the line in the direction it already goes. For example, (12, 96) is far right (high leverage) but changes the slope by less than 1%.

5 Apply a new situation, no tool

1. $\hat{y} = 15 + 2.4x$ predicts quiz score from hours of study. A student studied 5 hours and scored 31. Find and interpret the residual.
   $\hat{y} = 27$, residual $= 31 - 27 = 4$: she scored 4 points higher than predicted.
2. For a scatterplot, $r = -0.8$. What fraction of the variation in y does the linear model account for?
   $r^2 = 0.64$, so 64%.
3. Use the study model $\hat{y} = 51.97 + 3.706x$ to predict the score for 30 hours of study. Is the prediction trustworthy?
   ≈ 163 points: impossible. This is extrapolation far beyond the data (0–9 hours).

Teacher notes

Intended discovery: A residual plot with a pattern shows a line is the wrong model even when r is near 1; outliers have large residuals, high-leverage points have extreme x-values, and influential points change the line when removed; changing units rescales the slope but not r or r^2.

Likely misconceptions:

- A high r means a linear model is appropriate.
- Every outlier is influential, or every high-leverage point is influential.
- Interpreting r^2 as the percent of points on the line.

Answers that can vary: Explore 4 coordinates (many points work for each type).